

Multivariable Thresholding

Given the below grayscale image, partition it into 3 regions using 2 thresholds:

(a) Region 1: Dark with Intensities below 100

(b) Region 2: Medium with Intensities between 100 and 180.

(c) Region 3: Light with Intensities above 180.

34	120	87	205	65
155	76	134	90	182
30	199	45	170	150
220	101	123	180	95
66	140	85	160	210

① Define the thresholds:

Threshold 1 = 100

Threshold 2 = 180

② Segment the image:

If Intensity < 100 , classify as Region 1 (Dark)

If Intensity ≤ 180 , classify as Region 2 (Medium)

If Intensity > 180 , classify as Region 3 (Light)

③ Apply the segmentation to each pixel in the matrix.

④ The segmented image is shown below

1	2	1	3	1
2	1	2	1	3
1	3	1	2	2
3	2	2	2	1
1	2	1	2	3

1 $\rightarrow$ Region 1 (Dark)

2 $\rightarrow$ Region 2 (Medium)

3 $\rightarrow$ Region 3 (Light)